SECTION 13.4 - QUALITY ASSURANCE/QUALITY CONTROL PRACTICES FOR ORD LABORATORY AND FIELD-BASED RESEARCH

TABLE OF CONTENTS

LIST OF ACRONYMS 2
POLICY STATEMENT 2
REFERENCES 2
BACKGROUND 3
DEFINITIONS 3
Analytical Instrumentation 3
Facility Manual 4
Laboratory 4
Laboratory Records 4
Principal Investigator (PI) 4
Quality Assurance Project Plan 4
Quality Management Plan (QMP) 4
Standard Operating Procedure (SOP) 5
Supporting Equipment 5
RESPONSIBILITIES 5
Non-Federal On-site Employees 5
Principal Investigator and Other EPA Staff 5
Quality Assurance Manager 6
Supervisor 6
PROCEDURES 6
Laboratory Records 6
Standard Operating Procedures 6
Instrument and Equipment Calibration/Maintenance 6
Analytical Instrumentation 7
Supporting Equipment 7
Reagent Water Systems 8
Environmental Conditions 8
Laboratory Housekeeping 8
Demonstration of Proficiency 8
Analyst Proficiency 8
Field Staff Proficiency 9
Sample and Standard Storage 9
Standard Preparation 9
QA/QC Checks 10
Method Detection Limits (MDLs) 10
Data Review - Compliance with Acceptable Criteria 10
Data Storage 10
DELEGATIONS 10
APPROVAL 11
REVISION HISTORY 11
EXHIBIT 1: Requirements for Supporting Equipment Checks and Documentation 12

EXHIBIT 2: Supporting Equipment Checks and Documentation, Option to Use Documented Specialized Requirements 13
APPENDIX A: Training 15

LIST OF ACRONYMS

ANSI	American National Standards Institute
APPCD	Air Pollution Prevention and Control Division
CFR	Code of Federal Regulations
CRM	Certified Reference Material
IEC	International Electrotechnical Commission
ISO	International Organization for Standardization
LCA	Laboratory Competency Assessment
LOP	Laboratory Operating Procedure
MDL	Method Detection Limit
NIST	National Institute of Standards and Technology
ORD	Office of Research and Development
PI	Principal Investigator
PPM	Policy and Procedure Manual
QA	Quality Assurance
QC	Quality Control
QAM	Quality Assurance Manager
QAPP	Quality Assurance Project Plan
QMP	Quality Management Plan
SOP	Standard Operating Procedure
SRM	Standard Reference Materials
UV	Ultraviolet

POLICY STATEMENT

It is Office of Research and Development (ORD) policy that all laboratory and field-based research be conducted in a manner that meets or exceeds the basic quality assurance (QA)/quality control (QC) practices identified in this document. This policy addresses QA/QC minimum requirements for calibration, maintenance, and documentation for data generated to support ORD research. These QA/QC minimum requirements apply to both laboratory and field data collection activities.

This policy is intended for in-house ORD research and must be incorporated into the language for all extramural vehicles for work performed on site at EPA. Individual organizations have discretion in applying this policy to work performed under off site extramural vehicles.

REFERENCES

(1) EPA, Office of Environmental Information. *EPA QA Field Activities Procedure*; CIO 2105-P-02.0, 2014. http://www2.epa.gov/irmpoli8/epa-qa-field-activities-procedures.

(2) EPA Office of Research and Development (ORD). Policies and Procedures Manual, Section 13.2. *Scientific Recordkeeping: Paper*; current version.

(3) EPA Office of Research and Development (ORD). Policies and Procedures Manual, Section 13.6. *Scientific Recordkeeping: Electronic*; current version.

(4) EPA, ORD. ORD Policies and Procedures Manual, Chapter 13, Section 13.10. *External Laboratory Competency Assessments (LCAs)*; current version.

(5) EPA ORD. Policies and Procedures Manual, Section 13.14. *ORD Implementation of the EPA Field Activities Procedure*; current version.

(6) EPA Records Schedule 1035. http://intranet.epa.gov/records/schedule/final/1035.html.

(7) International Organization for Standardization/International Electrotechnical Commission (ISO/IEC) 17025. *General requirements for the competence of testing and calibration laboratories*; ISO/IEC: Geneva, Switzerland, 2005. Available from each organization's Quality Assurance Manager (QAM).

(8) U.S. Government Publishing Office Code of Federal Regulations, Title 40, Part 136, Appendix B. *Definition and Procedure for the Determination of the Method Detection Limit; 2011*. http://www.gpo.gov/fdsys/pkg/CFR-2011-title40-vol23/pdf/CFR-2011-title40-vol23-part136-appB.pdf.

BACKGROUND

Data generation and collection activities that are conducted in the laboratories and field by EPA ORD vary widely. Many of the practices in this policy apply to both laboratory and field data collection activities including calibration, maintenance, proficiency, documentation, and assessment. In addition, the processes of measurement or analysis and collection of data inherently possess a number of common features, no matter the methodology involved.

This policy provides basic required QA/QC practices for ORD units conducting laboratory or field research and related activities such as measurement or experimental set-up and collection of data in the field. More specific policies or Standard Operating Procedures (SOPs) with additional requirements may supersede this one where they exist and are applicable.

DEFINITIONS

Analytical Instrumentation

It is useful to introduce the category of analytical instrumentation for describing QA/QC practices typically associated with complex measurement systems. However, depending on use, QA/QC for analytical instrumentation may be similar to QA/QC for supporting equipment. Examples of analytical instruments are provided.

The term "Analytical Instrumentation" refers to devices including, but not limited to, gas chromatographs, gas chromatographs/mass spectrometers, inductively coupled plasma-optical emission or mass spectrometers, atomic absorption spectrophotometers, high performance liquid chromatographs, microscopes used for quantitative measures (e.g., scanning electron microscopes, transmission electron microscopes), gas proportional counters, gamma spectrophotometers, and meters for pH, conductivity, and dissolved oxygen. Field instruments include but are not limited to data sondes, various meters (e.g., anemometers, pH meters, conductivity meters), particle size analyzers, and portable spectrophotometers.

Facility Manual
A facility manual is an optional document that describes an EPA on-site facility and its capabilities, documents its management and staffing structure, personnel responsibilities, training requirements and records, provides operating procedures, and establishes basic recordkeeping requirements. The purpose of a facility manual is to document facility status, and facilitate maintenance and improvement of the efficiency and quality of operations. The facility manual is intended for internal use within the facility only. It is reviewed and approved by ORD facility management, quality assurance (in QA matters), and safety personnel. This is not a Quality Management Plan (QMP). In the event of an assessment, the information in the facility manual may be made available to the assessor to provide additional details about a facility's analytical capabilities and/or if needed to support QA practices.

Laboratory
An area equipped for scientists to make measurements.

Laboratory Records
"Laboratory records" is the collective term used for, but is not necessarily limited to, manuals, protocols, research notebooks, logbooks, compilations of data on pre-printed forms, instrument print-outs, and computer-resident notebooks and data that document the preparation, analysis, and results of data generation and collection activities. Laboratory records are maintained as per Record Schedule 1035 (6).

Principal Investigator (PI)
The PI is the person who is responsible for the technical leadership of the research on which he/she works, including the generation and execution of the quality assurance project plan (QAPP) as well as the implementation of this policy.

Quality Assurance Project Plan
A QAPP is a document describing in comprehensive detail the necessary quality assurance, quality control, and other technical activities that must be implemented to ensure that the results of the work performed will satisfy the stated performance criteria. In most cases QAPPs will be written for "tasks" described in ORD's research action plans or even lower aggregations of research (e.g., sub-task, research effort, etc.).

Quality Management Plan (QMP)
A QMP is a (planning) document that describes a quality system in terms of the organizational structure, policy and procedures, functional responsibilities of management and staff, lines of

authority, and required interfaces for those planning, implementing, documenting, and assessing all activities conducted.

Standard Operating Procedure (SOP)
SOPs are detailed descriptions of a repetitive function (e.g., analytical method, procedure, or technique) written to a level of detail that, when followed, allows the function to be performed in the same way with the same results between personnel and over time. Some ORD organizations may use variations on the name for the procedure, e.g., Operating Procedure, Research Operating Procedure, Laboratory Operating Procedure, Standard Operating Approach, and Miscellaneous Operating Procedure. However, the intent is the same, i.e., documenting the procedure used for an analytical method, procedure, or technique.

Supporting Equipment
It is useful to introduce the category of supporting equipment for describing a certain set of QA/QC practices. However, depending on use, QA/QC for supporting equipment may be similar to QA/QC for analytical instrumentation. Examples of supporting equipment are provided.

Supporting equipment includes but is not limited to balances and associated weights, pipettes, centrifuges, freezers, incubators, refrigerators, steam autoclaves, thermometers/thermocouples (working and reference), reagent water systems, thermocyclers, and ultraviolet (UV) lights (used to disinfect hard surfaces). See Exhibit 1 and Exhibit 2 for supporting equipment calibration and verification checks.

RESPONSIBILITIES

Non-Federal On-site Employees
On-site grantees, visiting scientists, students, and contractors that work in ORD laboratories or on ORD field studies are required to comply with this policy. Specific responsibilities include, but are not limited to, performing calibrations of analytical instrumentation, and documenting necessary user verification checks of equipment used for laboratory or field research activities. Contract procedures for on-site contractors must incorporate this requirement. For other parties, the mentor/host must ensure that these procedures are followed.

Principal Investigator and Other EPA Staff
The PI's responsibilities are limited to project management issues and as such the PI is responsible for ensuring that this policy is implemented for their projects. Specific responsibilities include ensuring that other on-site employees involved in the PI's projects are knowledgeable and trained to fulfill their daily responsibilities with respect to the maintenance, calibration, and user verification requirements of this procedure for equipment and instrumentation used in the project. PIs are also responsible for participating in audits and surveillances, and for ensuring any correction actions are implemented.

Other EPA staff are responsible for implementing this policy and fulfilling their daily responsibilities as noted above.

Quality Assurance Manager
The QAM or designee shall conduct an internal audit (using this policy, PPM 13.4, as the audit standard) for each laboratory (see definition) in their respective divisions on a schedule such that all laboratories (including contractor operated laboratories) are assessed at least once every three-year period.

The QAM or designee is responsible for coordinating calibration services for supporting equipment in the requirement timeframes and maintaining records of these services.

Supervisor
The supervisor is responsible for ensuring that all personnel in his/her organizational unit are aware of the requirements of this policy and that it is implemented rigorously and thoroughly (e.g. by supporting internal audits and necessary correction actions). Employee performance issues, as related to this policy, are the responsibility of the supervisor.

PROCEDURES

Laboratory Records
Research activities shall be sufficiently documented to allow reconstruction of those activities by someone of comparable education and training. Documentation of activities can be recorded in research notebooks, electronic files, logbooks, and other appropriate media. Refer to PPM 13.2 (2) and PPM 13.6 (3) for further discussion of laboratory recordkeeping requirements.

Standard Operating Procedures
If a standard or citable procedure is not explicitly followed, such as an EPA method or ASTM International Standard, SOPs shall be prepared to document analytical procedures that will be performed routinely or that will be implemented by multiple personnel. Other situations may also warrant the generation of an SOP (e.g. to allow more flexibility in the method). SOPs or other documented standard procedures shall be available in the laboratory or field area where the procedure is implemented. Existing SOPs shall be reviewed every two years to ensure that they reflect current practice, and these reviews shall be documented.

Instrument and Equipment Calibration/Maintenance
All activities discussed in this section shall be documented following the requirements of PPM 13.2 (2) and PPM 13.6 (3), and include identification of the equipment or instrument, user, date, activity performed, and results or observations. All analytical instruments shall be properly maintained and calibrated on a regularly scheduled basis with documentation of these activities. Documentation may be made in paper or electronic laboratory records maintained by the PI or other employee (for project-dedicated instruments), or in instrument logbooks maintained in the laboratory or field (for shared instruments). Instrument logbooks shall be kept in a readily accessible location near the instrument. For calibrations that are managed centrally by the QAM (e.g., balances, pipettes), documentation shall be maintained in the QA files. Vendors that provide calibration services shall be accredited to International Organization for Standardization/International Electrotechnical Commission (ISO/IEC) 17025 (7) where available. For calibration services received by non-ISO/IEC 17025 accredited vendors, sufficient justification must be made, documented, and authorized by the organization's management.

Alternatively, in-house metrology laboratories that provide calibration services for ORD support equipment may be used; however, these laboratories shall follow the requirements of this PPM and are subject to the internal and external laboratory competency audits as described in PPM 13.10 (4).

Analytical Instrumentation

Laboratory and field analytical instrumentation shall, if possible, be calibrated prior to use with standards of known and documented quality and uncertainty, traceable to a recognized standard organization whenever available. In the event that calibration standards are not available commercially, they may be prepared in-house in accordance with written procedures that have been reviewed and approved. A verification of the initial calibration should be conducted using a standard from a different source (second source) than that used for the initial calibration. Examples of second sources are Standard Reference Materials (SRMs), Certified Reference Materials (CRMs), performance evaluation samples, a standard from a vendor other than the calibration standard vendor or from a different lot from the same vendor. Acceptance criteria shall be specified for second source standards in the SOP/QAPP/facility manual/research notebook, and corrective actions documented in instances where criteria are not met. If this second source check is not performed, justification, such as lack of availability, shall be documented in a SOP/QAPP/facility manual. Calibration shall be checked at the beginning and end of a sample set and periodically during use, according to manufacturer's instructions or as otherwise documented in an SOP or method. The calibration range of an instrument should not be exceeded. Results reported outside calibration ranges shall be qualified to reflect the fact that the accuracy of the reported concentration is uncertain.

Supporting Equipment

Prior to project initiation, the PI or a designee shall ensure that supporting equipment is in good operating condition and is properly calibrated, if the equipment can be calibrated. Appropriate QA/QC checks should be performed as presented in Exhibit 1 and Exhibit 2.

Balance/Weights Maintenance and Calibration

Balances (analytical, micro-, top-loading, etc.) shall be serviced annually and have their calibrations re-certified by an outside vendor or in-house metrology laboratory against American National Standards Institute/ASTM (ANSI) Class 1 weights (e.g., National Institute for Standards and Technology (NIST) traceable reference weights) that encompass the range of use. Balance calibration also shall be checked periodically by the user, or designee, with secondary "working" weights of an appropriate ANSI/ASTM classification as determined by the accuracy required for the research and that bound the range of use. These calibration checks are recommended once each day the balance is in use, unless justification for less frequent checking is documented in a QAPP (or similar planning document), SOP, or facility manual. Calibration checks shall be documented in paper or electronic laboratory records for the project or in instrument logbooks for the laboratory or field unit. The working weights are verified against ANSI/ASTM Class 1 traceable reference weights on a regularly scheduled, annual basis. In order to maintain traceability to national standards, the ANSI/ASTM Class 1 traceable reference weights shall be certified on a regularly scheduled basis by an ISO/IEC 17025 (7) accredited organization, at a frequency of at least once every five years.

Pipette Maintenance and Calibration

Mechanical variable and fixed volume pipetting devices shall be subjected annually to routine maintenance and calibration by an ISO/IEC 17025 (7) accredited outside vendor or in-house metrology lab to ensure that expected volumes are delivered within acceptance criteria as determined by the nature and frequency of use of the pipette. If volume dispensing issues are suspected, the pipette should be removed from service until a verification of performance is conducted.

Reagent Water Systems

Single-lab and common-use reagent water systems should be maintained according to manufacturer's specifications. Maintenance of the system should be documented in a logbook, but additionally may be documented by an affixed label on the system itself. User verification checks include reading and documenting resistivity meter display weekly with use to ensure water quality meets system specifications and user requirements. User verification checks shall be documented in paper or electronic laboratory records for the project or in instrument logbooks for the laboratory.

Environmental Conditions

Facility managers typically monitor a normal set of environmental conditions. In addition, if temperature, pressure, humidity, atmospheric composition, or any other environmental condition must be controlled during the implementation of analytical or field procedures (e.g., incubator temperature for microbiological tests), it shall be monitored and documented to ensure that the required range of the environmental condition is maintained. Monitoring instrumentation should be calibrated, if applicable, or verified at a specified frequency using traceable calibration standards of known and documented uncertainty.

Laboratory Housekeeping

Laboratory housekeeping shall be maintained to prevent contamination of samples and experiments. Floors, counters, and cabinets shall be kept free of trash and extraneous paper to avoid dust. Laboratory benches shall be decontaminated as needed.

Demonstration of Proficiency

The *ORD Certification Statement for Demonstration of Capability Form* is an example of a form used to document proficiency. The completed form is placed in the project file, if related to a specific project. If not project related, then the Supervisor or designee shall maintain documentation of these activities.

Analyst Proficiency

Analysts with no prior experience with a technique/method/instrument shall demonstrate and document their proficiency before analyzing samples. They must demonstrate proficiency by: 1) performing valid initial calibrations, 2) performing method detection limit determination, if appropriate, 3) demonstrating that his/her results meet all minimum QA/QC acceptance criteria as presented in the method document, e.g., the SOP, and if available, 4) satisfactorily analyzing a performance evaluation sample or, if unavailable, a second source standard. Documents generated to support demonstrations of proficiency are records that should be saved and readily associated with the ORD training certification statement, as appropriate.

Field Staff Proficiency

Prior to conducting field activities, the PI shall work with first-line supervisors to ensure that all staff assigned to the project are proficient in sampling protocols and equipment associated with the project. For additional details, refer to the Quality Assurance Field Activities Procedure (1) and PPM 13.14 (5). Documents generated to support demonstrations of proficiency are records that should be saved and readily associated with the ORD training certification statement, as appropriate.

Sample and Standard Storage

Refrigerators, refrigerated rooms, and freezers used to store samples shall be monitored for temperature. It is recognized that frequency of monitoring can be dependent on available equipment and can range from continuous to a particular moment in time when the user reads a thermometer. At a minimum, the frequency of monitoring will be weekly, unless otherwise documented in a QAPP or SOP. Acceptance criteria shall be defined and documented in an organization's QMP, policy, or SOP for a specific storage area, project or organization, as appropriate. If the temperature falls outside of established acceptance criteria, corrective action should be taken and documented and the potential impact on samples evaluated.

Samples and standards shall be maintained to ensure their integrity. Integrity can be demonstrated using requirements from standard methods or literature, or by empirical demonstration. For example, based on available published methods, a project-specific QAPP or a SOP may specify that samples require refrigeration if sample analysis is not to be performed immediately upon sample collection or sample preparation. Any sample or standard not consumed during the analytical procedure must be immediately returned to the refrigerator or freezer (or other comparable environmental conditions) until it is determined that additional analyses are not needed.

Adequate measures shall be taken to protect stored samples and standards from each other, heavily contaminated samples, and other materials that could potentially cross-contaminate them. For example, this could be accomplished by appropriate secondary containment within a single refrigerator or freezer.

Applicable sample holding times and preservation requirements shall be followed and documented in project-specific QAPPs/SOPs and based on available published methods in order to minimize the loss of analytes of interest.

Standard Preparation

Standard preparation documentation shall include as appropriate the date, analyst identification, the identity of the stock/intermediate standard used and its source/lot number, identifying chemical information, preparation procedures, concentrations, solvent, carrier or buffer, and applicable expiration dates. The documentation shall be recorded in paper or electronic laboratory records for the project or in instrument logbooks for the laboratory. Standards shall be stored to maintain their integrity. Researchers shall be responsible for maintaining the integrity of standards, or disposing of them on a timely basis if they have exceeded their expiration date.

QA/QC Checks
QA/QC checks of analytical procedures, methods and techniques shall be performed as defined in applicable SOPs/QAPPs/facility manuals. The results of the QA/QC checks shall be recorded in paper or electronic laboratory records for the project or in instrument logbooks for the laboratory. QA/QC checks for field procedures such as field blanks and duplicates will be described in QA Project Plans and depend on the needs of the project.

Acceptance criteria for QA/QC checks may be documented in SOPs/QAPPs/facility manuals. When QA/QC checks are not defined in SOPs/QAPPs/facility manuals, at a minimum and as possible, positive controls (e.g., standards of known composition or matrix spikes), negative controls (e.g., blanks), and replicates (e.g., duplicates) should be performed periodically where applicable to demonstrate the accuracy and precision of a method for each unique matrix.

When available, the uses of SRMs or CRMs are strongly encouraged to provide for additional validation of the analytical process. Cost considerations might dictate that they be used only periodically.

Method Detection Limits (MDLs)
It is understood that a variety of techniques as well as terminology surround the issue of detection and quantitation limits, and therefore the method for determination of the limits shall be documented for instruments that report values below the lowest calibration standard. One approach is found in Title 40 Code of Federal Regulations (CFR) Part 136 Appendix B (8). MDLs shall be determined if results beyond the lowest and highest calibration standards are reported. Results beyond calibration ranges shall be qualified to reflect the fact that the accuracy of the reported concentration is uncertain. The concentration of the lowest (or highest) calibration standard should also be provided with the sample data so that the user of the data has as much information as possible about any concentration outside the instrument calibration range.

Data Review - Compliance with Acceptable Criteria
For each analysis, the analyst shall review results for QC checks performed to determine compliance with acceptance criteria specified in the applicable SOPs/QAPPs/facility manuals. If acceptance criteria are not met, the analyst shall perform corrective action as required by the applicable SOPs/QAPPs/facility manuals. The results of the data review shall be recorded in paper or electronic laboratory records for the project, and shall be provided to QAMs, if requested, during their review of data related to publications.

Data Storage
Documentation of research activities, including all analytical data, (electronic and/or hardcopy) shall be maintained as required by EPA's records management policies. The pertinent EPA Records Schedules for ORD research records is 1035 (6).

DELEGATIONS
Not Applicable

APPROVAL

/s/ Jerry Blancato March 30, 2017
Senior Information Official for Research and Development Date

Effective October 1, 2017

Minor revision July 8, 2021

EXHIBIT 1: Requirements for Supporting Equipment Checks and Documentation

Equipment	Calibration/ Maintenance Frequency Notes*	User Verification Checks**	Required Documentation
Balances	Annually	Daily with use	User verification checks kept on log sheets or logbooks, or associated research notebook if done for a specific project and maintained by on-site employees. Annual calibration records maintained by QAM or other designated party.
Pipettes	Annually	Verification recommended approximately 6 months after annual calibration.	Annual calibration records maintained by QAM or other designated party. User verification checks kept on log sheets or logbooks, or associated research notebook if done for a specific project.
Temperature monitoring, automated (Refrigerators & Freezers)	Annually	None	Logbooks for each probe are not required. Annual calibration records maintained by QAM or other designated party.
Temperature monitoring, manual (Refrigerators & Freezers)	See "Thermometers, working"	Temperatures documented at least weekly unless otherwise documented in QAPP or SOP.	Log sheets or logbooks kept for each refrigerator/freezer. Also see "Thermometers, working"
Thermometers, reference	Verified once every 5 years using an ASTM/NIST thermometer	None	Verification records maintained by QAM or other designated party.
Thermometers, working	None	Verified annually using reference thermometer	Verification records maintained by QAM or other designated party
Weights, traceable reference	Once every 5 years	None	Certification records maintained by QAM or other designated party.
Weights, working	Verified annually using traceable reference weights.	None	Verification records maintained by QAM or other designated party.

*****Annual calibrations or verifications should be indicated by a sticker attached to the equipment. Contracted vendors with ISO/IEC 17025 (7) (or similar) accreditation or in-house metrology labs may perform these calibrations.
**User verification checks apply to equipment in active use. Out-of-use equipment shall be clearly labeled as such and documented in the associated logsheet/logbook. Return to active use, shall also be indicated in logsheet/logbook.

EXHIBIT 2: Supporting Equipment Checks and Documentation, Option to Use Documented Specialized Requirements***

Equipment	Calibration/ Maintenance Frequency Notes*	User Verification Checks**	Required Documentation
Autoclaves, steam	Annually (time, temperature, and pressure)	Biological indicator-monthly spore checks. Chemical indicator/integrator-with each use. Mechanical monitoring (temperature & pressure gauges and recording charts) with each use.	All calibration/maintenance records kept by responsible technical staff or other designated party. User verification checks kept on log sheets or in logbooks.
Centrifuges (microcentrifuges exempted)	Annually (e.g. rotational velocity, temperature, timer, brushes, rotors, as appropriate)	None	Annual calibration records maintained by responsible technical staff or other designated party.
Incubators	CO_2 flow calibrated annually.	Temperatures and gas concentration (if applicable) documented daily with use using internal thermometer. Biweekly checks of CO_2 concentration with the Fyrite Analyzer with use.	User verification checks kept on log sheets or logbooks, or associated research notebook if done for a specific project and maintained by on-site employees. All calibration/maintenance records kept by responsible technical staff or other designated party.
Reagent water systems (single-lab)	None. Systems maintained according to manufacturer's specifications.	Resistivity meter readout documented weekly with use to ensure water quality meets system specifications.	Log sheets or logbooks kept for each unit.
Reagent water systems (facility-wide and centrally managed)	None. Systems maintained according to facility specifications	None. Rely on central monitoring and logging of water quality.	Facility logs
Thermocyclers	Annually (thermal characteristics that influence the outcome of a polymerase chain	None	Calibration records maintained by user or party responsible for instrument. Diagnostic results recorded in research records.

Equipment	Calibration/ Maintenance Frequency Notes*	User Verification Checks**	Required Documentation
	reaction, including accuracy, uniformity, overshoot/ undershoot, heat/cool rate and hold time)		
UV lights	None	Verification is performed yearly to ensure lights are meeting minimum specification for UVC flux.	Verification records maintained by user.

*Annual calibrations or verifications should be indicated by a sticker attached to the equipment. Contracted vendors with ISO/IEC 17025 (7) (or similar) accreditation or in-house metrology labs may perform these calibrations.
**User verification checks apply to equipment in active use. Out-of-use equipment shall be clearly labeled as such and documented in the associated logsheet/logbook. Return to active use, shall also be indicated in logsheet/logbook.
*** Specialized documented requirements that are different from those listed should be documented in QAPPs and/or SOPs.

APPENDIX A: Training

Chapter 13 of the ORD Policies and Procedures Manual focuses on quality assurance (QA) for ORD's research. Training for Chapter 13 is made available in several ways, as follows:

- One-on-one training with a QA Manager.
- Self-directed training using Power Point slides and/or recorded webinars are available on ORD@Work.
- Group classroom style training, available by request.
- Group training via webinar which is scheduled for newly approved sections of Chapter 13.

Training for QA Managers (QAMs) by ORD senior QA staff will precede delivery of training by QAMs to research staff.

Questions about training can be directed to any member of ORD's QA staff.